

# Research Idea: NGC4429

ALMA Thesis Planner (automated pipeline)

**Title: Dense gas around the supermassive black hole of NGC 4429: an HCN(1-0) and 3 mm continuum characterization of the circumnuclear disk**

**Core question:** How much dense gas, as traced by HCN(1-0), resides in the circumnuclear disk of NGC 4429, how is it distributed at the delivered  $\sim 1.9''$  ( $\sim 150$  pc at Virgo distance) resolution, and what does that imply — quantitatively and with honestly stated limits — for the proposal’s question of whether the known central molecular hole is truly gas-devoid or merely faint in one tracer?

**Framing decisions made upfront:** (a) A simple beam-versus-hole comparison ( $1.9''$  beam  $\approx 150$  pc versus a  $\sim 10$ – $100$  pc cavity) shows the cavity is sub-beam, so no cavity measurement is attempted as a centerpiece; the cavity question lives in one compact, clearly bounded constraint section. (b) The thesis does not assume any specific external dataset exists in usable form. Any use of literature values (disk geometry, CO fluxes) is treated as contingent: each has a pre-committed fallback that keeps the thesis complete using only the two delivered FITS products, and the write-up states which path was taken and why.

**Plan of work, using only the two delivered FITS products:**

1. **Cube hygiene and the week-2 gate.** Measure the per-channel RMS, verify beam and astrometry against the FITS headers, and extract the spatially integrated spectrum at the galaxy position. A pre-registered criterion (integrated line flux at  $\geq 5\sigma$ , or per-beam peak S/N  $\geq 3$  in  $\geq 3$  contiguous channels) selects exactly one branch. Tooling for the unselected branch is not built — the gate decides before any branch-specific code is written, so no month is spent on machinery half of which is discarded. A few days of quick convolution checks (toy filled versus holed disk convolved with the actual beam) confirm the expected detectability floor for any central deficit and are recorded as an appendix figure.

2. **Detection branch — spatial and spectral characterization.** Moment 0 map with masking appropriate to the  $\sim 49.6$  km/s channels; moment 1 only if the line spans enough channels (moment 2 excluded as unreliable at this channel width). Total HCN(1-0) flux, line width, and systemic velocity. A major-axis position–velocity diagram is included strictly as a sanity check that the emission is disk-like — at a  $\sim 150$  pc beam and  $\sim 50$  km/s channels across a disk only a few beams wide, it is explicitly not presented as a kinematic result, and no SMBH mass modeling is attempted.

3. **Non-detection branch — pre-committed to be a full thesis, not a stub.** (a) The primary limit method is deliberately self-contained: rigorous per-pixel and per-aperture upper-limit maps, plus a plain multi-aperture spectral stack (central beam, inner disk, full disk annuli) over the line window, which at  $\sim 49.6$  km/s channels captures nearly all the sensitivity gain available. (b) Velocity-coherent shift-and-stack using a literature rotation model is a stretch goal only, attempted if a published disk geometry (systemic velocity, PA, inclination) is straightforwardly available —

since rotation spreads emission over only  $\sim 1\text{--}3$  channels across much of the disk, its expected gain over the plain stack is modest and the thesis does not depend on it. (c) Dense-gas-fraction limits are likewise contingent: they are computed only if a published CO flux can be defensibly matched to the HCN aperture and tracer resolution — and that matching is budgeted as real analysis work, not a lookup. The pre-committed fallback, stated in the thesis plan from day one, is to report HCN(1-0) flux and mass limits alone, which remain a complete and defensible result for a dataset targeting exactly this question.

**4. Central-deficit constraint envelope (short, appendix-adjacent section in either branch).**

Parametric filled-disk models with and without central cavities, convolved with the true beam and compared to the observed central-beam flux and radial profile. The disk-profile assumption is bracketed, not fixed: exponential and Gaussian profiles over a plausible range of scale lengths, with the reported exclusion region being the conservative intersection and anything assumption-dependent labeled unconstrained. Expected honest headline: "cavities below  $\sim Y$  pc or shallower than  $Z\%$  are unconstrained at this resolution; only cavities of order the beam and near-total in depth would be excluded/detected."

**5. Continuum and mass budget.** Nuclear 3 mm continuum position and flux (or upper limit) from the wideband image, as context for the AGN-versus-secular origin discussion. HCN flux or limit converted to dense-gas mass (central beam and full disk) using standard conversion factors, with the factor-of-several systematic uncertainty propagated explicitly so the mass budget is a bracketed range.

**Why this is thesis-scoped:** two science-ready FITS files, Python-only analysis (astropy/spectral-cube-style tools, aperture photometry, spectral stacking, beam convolution of toy models), one executed branch selected by a week-2 gate, and no step whose completion depends on an external dataset the description does not provide — every literature-dependent element carries an explicit, delivered-data-only fallback. The CO(1-0) and CO(6-5) lines named in the proposal abstract are not among the delivered products and no substitute dataset is named in the description, so the thesis is built on HCN(1-0) and the Band 3 continuum alone; the CO(2-1) hole and any published disk properties enter only as motivating context or optional comparison values, never as data the thesis requires.